

Remedies for the Force-Magnitude Dip in an Inverse-Elliptical Potential Field

Abstract

A companion to *Why an Inverse-Elliptical Potential Produces a Force-Magnitude Dip Along its Major Axis*, which established that $|\nabla U|$ has a strict local minimum on the major axis whenever the elongation q exceeds $\sqrt{3}$. Here we identify precisely which factor of the gradient carries the defect, and evaluate four candidate remedies against two requirements: the dip must be removed, and the force *direction* must be preserved so that the tangential component driving lateral evasion survives. We show that normalising the gradient by $|\nabla \rho|$ is the only candidate meeting both, prove it removes the dip for every q , and quantify the price paid, namely loss of conservativeness. A decision rule in terms of q concludes.

1 What is and is not the defect

It is worth first excluding a plausible but incorrect diagnosis. On the major axis one has $F_x \equiv 0$ whenever the obstacle is laterally symmetric, since $U(x, y) = U(-x, y)$ implies $\partial_x U(0, y) = 0$ identically. This is a symmetry statement, independent of q , and it holds equally for the *circular* field $q = 1$, which exhibits no dip at all. Vanishing lateral force on the axis is therefore not the phenomenon under study and cannot be what distinguishes the elliptical case.

The genuine distinction is the sign of the second derivative, i.e. the character of the critical point. Along a horizontal scan at height s :

- for $q < \sqrt{3}$ (including the circle), $x = 0$ is a *maximum* of $|F|$, which then decays monotonically outward;
- for $q > \sqrt{3}$, $x = 0$ is a *minimum*, and $|F|$ first rises to an off-axis ridge before decaying.

The force is weakest exactly where the obstacle lies most directly ahead. The practical consequence appears whenever $|F|$ is thresholded as a scalar risk measure: at $q = 12$, $s = 9.5$ the on-axis value is 4.80 while the flank ridge reaches 12.12, so a trigger set at 8 fires on the flanks but not straight ahead.

Remark 1 (The dip is relative, not absolute). *Elongation does not weaken the on-axis force. With $t_x = 1$ and $q = t_y/t_x$, the elliptical radius on the axis is $\rho(0, s) = s/t_y$, so $U(0, s) = \lambda t_y/(s - l)$ and*

$$|F_y(0, s)| = \frac{\lambda t_y}{(s - l)^2}, \quad (1)$$

which increases linearly with t_y ; numerically 1.200, 2.000, 4.800 for $q = 3, 5, 12$ at $s = 9.5$, $\lambda = 10$, $l = 4.5$. The ridge merely grows faster, as $O(t_y^2)$. The dip is thus a purely relative feature, and any remedy should be judged on restoring monotonicity, not on raising the on-axis magnitude.

2 Localising the defect

Let

$$\rho = \sqrt{x^2 + \frac{y^2}{q^2}}, \quad U = U(\rho), \quad (2)$$

so that U depends on position only through ρ . By the chain rule $\nabla U = U'(\rho)\nabla\rho$, hence

$$|\nabla U| = \underbrace{|U'(\rho)|}_{\text{function of } \rho \text{ alone}} \cdot \underbrace{|\nabla\rho|}_{\text{varies around the ellipse}}. \quad (3)$$

Proposition 2 (The first factor is harmless). *At fixed height $s > 0$, $\rho(x, s)$ is strictly increasing in $|x|$. Hence if U' is of one sign and $|U'|$ is monotone in ρ , the factor $|U'(\rho)|$ is monotone in $|x|$ with its extremum at $x = 0$, and can never produce an interior ridge.*

Proof. $\partial_x \rho = x/\rho > 0$ for $x > 0$, so ρ strictly increases with $|x|$; composing a monotone function of ρ with a monotone function of $|x|$ preserves monotonicity. $\square$

The whole defect therefore resides in $|\nabla\rho|$. Explicitly,

$$|\nabla\rho| = \frac{\sqrt{x^2 + y^2/q^4}}{\rho}, \quad |\nabla\rho| = \begin{cases} 1/q & \text{on the major axis } (x = 0), \\ 1 & \text{on the minor axis } (y = 0). \end{cases} \quad (4)$$

This factor sweeps the range $[1/q, 1]$ as one moves around an ellipse, and it is precisely this q -fold swing that overturns the decay of $|U'|$ once $q^2 > 3$. Equation (4) immediately suggests the two lines of attack pursued below: *shrink the range* of the offending factor (reduce q), or *divide it out*.

3 Candidate remedies

We evaluate each candidate against two requirements:

- (R1) the contrast $C = |F|_{\max}/|F|_{x=0}$ on a horizontal scan equals 1, i.e. the dip is removed;
- (R2) the force *direction* is preserved, so the tangential component that drives lateral evasion is not destroyed.

Requirement (R2) is not cosmetic. A remedy that leaves the force purely radial reproduces the classical artificial-potential-field deadlock, in which a vehicle directly behind an obstacle receives braking but no lateral guidance.

3.1 Remedy A: reduce the elongation q

By Theorem 1 of the companion note, the dip vanishes outright for $q \leq \sqrt{3}$. In the implemented field the effective threshold is more forgiving, because the vehicle-dimension prefactor $(1 - l/s)^{-1}$ contributes an additional x -dependent term to F_y that partially fills the minimum. Measured on the implemented form:

$q = t_y/t_x$	C , pure ellipse	C at $s = 7.5$	C at $s = 9.5$
12	4.651	1.982	2.524
5	2.005	1.091	1.241
3	1.299	1.000	1.007
2	1.026	1.000	1.000
$\sqrt{3}$	1.000	1.000	1.000
1	1.000	1.000	1.000

Thus $q \leq 3$ already satisfies (R1) to three decimal places in practice. Requirement (R2) is met trivially, the field being unmodified. Remedy A preserves conservativeness and retains U as a candidate Lyapunov function; its only cost is that q is a modelling parameter, so lowering it narrows the longitudinal reach of the danger zone.

3.2 Remedy B: steepen the radial power

Generalise to $U = \rho^{-n}$. Then $\nabla U = -n\rho^{-n-2}(x, y/q^2)$ and, with $u = x^2$ and $a = s/q$,

$$g(u) = |\nabla U|^2|_{y=s} = n^2 \frac{u + a^2/q^2}{(u + a^2)^{n+2}}. \quad (5)$$

Differentiating and cancelling $(u + a^2)^{n+1}$ as before,

$$\text{sign } g'(u) = \text{sign} \left[a^2 \left(1 - \frac{n+2}{q^2} \right) - (n+1)u \right], \quad (6)$$

so the dip criterion generalises to

$$\boxed{q > \sqrt{n+2}} \quad (7)$$

recovering $\sqrt{3}$ at $n = 1$. Raising n does raise the threshold, but far too slowly to be useful, and the response is not even monotone in n :

n	$\sqrt{n+2}$	C at $q = 5$	C at $q = 12$
1	1.73	1.241	2.524
2	2.00	1.302	2.760
3	2.24	1.280	2.712
6	2.83	1.170	2.390
10	3.46	1.074	2.062
23	5.00	1.000	1.549

Suppressing the dip at $q = 5$ requires $n \approx 23$, and $n = 2$ is worse than $n = 1$. Remedy B is rejected.

3.3 Remedy C: the metric-consistent force

Take $F = -G^{-1}\nabla U$ with $G = \text{diag}(1, 1/q^2)$, the metric induced by ρ . For $U = 1/\rho$ this gives $F = (x, y)\rho^{-3}$, so $|F| = r\rho^{-3}$ with $r = \sqrt{x^2 + y^2}$, and

$$g(u) = \frac{u + s^2}{(u + a^2)^3}, \quad \text{sign } g'(u) = \text{sign} [a^2 - 3s^2 - 2u] < 0 \quad (8)$$

for all $u \geq 0$ whenever $a < s\sqrt{3}$, i.e. $q > 1/\sqrt{3}$. Requirement (R1) is met for every practical q .

Requirement (R2), however, fails. The force becomes exactly radial, and the measured force angle at $x = 0.5$, $s = 9.5$ collapses to 1.6° , 1.5° , 1.3° , 0.7° for $q = 3, 5, 12, 30$ — against 13.8° , 33.8° , 73.4° , 85.0° for the unmodified field. Essentially all lateral guidance is destroyed. Remedy C is rejected.

3.4 Remedy D: normalise by $|\nabla\rho|$

Equation (3) localised the defect in $|\nabla\rho|$; Remedy D simply divides it out. Define

$$F = -\frac{\nabla U}{|\nabla\rho|} = -U'(\rho)\hat{n}, \quad \hat{n} = \frac{\nabla\rho}{|\nabla\rho|}. \quad (9)$$

Theorem 3 (Remedy D removes the dip at every elongation). *For F as in (9) with $U = 1/\rho$, and for every $q > 0$ and every height $s > 0$, the magnitude $|F|$ is strictly decreasing in $|x|$ and attains its maximum at $x = 0$. In particular $C = 1$, and no off-axis ridge exists.*

Proof. Since $\hat{n}$ is a unit vector, $|F| = |U'(\rho)| = \rho^{-2}$, a function of ρ alone and strictly decreasing in ρ . At fixed s , $\rho(x, s) = \sqrt{x^2 + (s/q)^2}$ is strictly increasing in $|x|$ by Proposition 2. The composition of a strictly decreasing with a strictly increasing function is strictly decreasing, so $|F|$ attains its maximum at $x = 0$ and has no interior critical point. No hypothesis on q was used. $\square$

Corollary 4 (Direction is untouched). *Dividing by the strictly positive scalar $|\nabla\rho|$ rescales the vector without rotating it, so F in (9) is everywhere parallel to $-\nabla U$. Requirement (R2) holds exactly.*

Both claims are confirmed numerically at $s = 9.5$. The angle column is identical to that of the unmodified field, and the contrast is exactly unity:

mode	q	$ F (0)$	peak	x^*	C	angle at $x = 0.5$
plain	3	1.200	1.208	0.828	1.007	13.8°
plain	5	2.000	2.482	1.070	1.241	33.8°
plain	12	4.800	12.117	0.540	2.524	73.4°
plain	30	12.000	73.376	0.223	6.115	85.0°
norm	3	3.600	3.600	0	1.000	13.8°
norm	5	10.000	10.000	0	1.000	33.8°
norm	12	57.600	57.600	0	1.000	73.4°
norm	30	360.000	360.000	0	1.000	85.0°
metric	3	10.800	10.800	0	1.000	1.6°
metric	5	50.000	50.000	0	1.000	1.5°
metric	12	691.200	691.200	0	1.000	1.3°
metric	30	10800.000	10800.000	0	1.000	0.7°

4 The price of Remedy D

Scale. Since $|\nabla\rho| \rightarrow 1/q$ on the axis, the normalisation inflates magnitudes by up to a factor q there ($4.800 \rightarrow 57.600$ at $q = 12$). If absolute force values are calibrated, rescale λ by $O(1/q)$.

Loss of conservativeness. Write $F = -\Phi \nabla \rho$ with $\Phi = U'(\rho)/|\nabla \rho|$. A field of this form is a gradient if and only if Φ is a function of ρ alone. But by (4), $|\nabla \rho|$ varies from $1/q$ to 1 *along a single ellipse* $\rho = \text{const}$, so Φ is not such a function and F is not conservative for any $q > 1$. Consequently no scalar potential has F as its descent direction, and U is no longer available as a Lyapunov function.

The magnitude of the violation, measured as $(\nabla \times F)/|F|$ at $(x, y - y_o) = (1.0, 7.5)$, is

q	$(\nabla \times F)/ F \text{ (m}^{-1}\text{)}$	$ F $
1	0.0000	1.095
3	-0.4384	6.603
5	-0.4789	10.163
12	-0.1225	18.907
30	-0.0203	24.095

The vanishing entry at $q = 1$ is a consistency check: for a circle $|\nabla \rho| \equiv 1$, so Remedy D reduces to the identity. Note the violation is *non-monotone*, peaking near $q \approx 5$ and becoming mild at large q — fortunately, the regime in which Remedy D is actually needed.

5 Decision rule

Remedy	(R1)	(R2)	conservative	verdict
A. reduce q to ≤ 3	yes	yes	yes	preferred when q may be lowered
B. steepen power n	no	yes	yes	rejected: needs $n \approx 23$
C. metric gradient	yes	no	yes	rejected: radial, causes deadlock
D. normalise by $ \nabla \rho $	yes	yes	no	preferred when large q is required

Concretely:

- **If $q \lesssim 5$:** apply Remedy A. At $q = 5$ the contrast is already only 1.241, and at $q = 3$ it is 1.007 — the dip is gone for practical purposes, at no structural cost.
- **If $q \gtrsim 12$ is required by the model:** apply Remedy D. It is the only candidate satisfying both (R1) and (R2), and its circulation penalty is mildest precisely in that regime (0.12 m^{-1} at $q = 12$ versus 0.48 m^{-1} at $q = 5$).

Finally, a scale check that may render the question moot. The ridge sits at $x^* \approx s/(q\sqrt{2})$, so at $q = 12$, $s = 9.5$ one finds $x^* = 0.554 \text{ m}$, which is smaller than the half-width 0.875 m of a 1.75 m vehicle. The entire low-force channel is then narrower than the host vehicle itself: a vehicle centred on the axis already has its corners in the high-force ridge. At such elongations the dip is largely an artefact of evaluating a point field, and no remedy may be warranted at all.